

# Amitabha Deb

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## Education

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### Master of Science, Robotics

University of Minnesota, Twin Cities

Aug 2022 - May 2024

GPA: 3.96 / 4.00

*Subjects:* Intelligent Robotic Systems, Computer Vision, Deep Learning, Probabilistic Reasoning and Reinforcement Learning, Machine Learning, Artificial Intelligence, Natural Language Processing

### Bachelor of Technology, Electronics and Communication

National Institute of Technology, Silchar

Aug 2015 - May 2019

GPA: 8.13 / 10.00

*Subjects:* Control Systems, Data Communication and Networks, Data Structures, Neural Networks and Fuzzy logic

## Work Experience

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### Applications Engineer II - AutoCAD, AGVs, Kollmorgen Application Toolkit

Remote, USA

Toyota Material Handling Inc.

Oct 2024 – present

- Designed system architectures for Raymond Autonomous Guided Vehicle (AGV) fleet deployments, delivering customized warehouse automation solutions.
- Analyzed fleet performance and operational data to improve system efficiency, define new product features and drive continuous system enhancements.
- Established system design standards, best practices, and technical documentation to improve solution consistency, scalability, and cross-functional collaboration.

### Software Engineer - Java, SQL, Windows Server, React, Spring Boot

Bengaluru, India

Lowe's

July 2019 – Aug 2020

- Handled the functionality of the Payment Switch enabling 19M transactions per week, ensuring real-time payment processing.
- Upgraded the encryption for store-to-payment switch data flow as a part of the End-to-end encryption project, enhancing data security.
- Enhanced Receipt Retrieval and Chargeback Management System applications, improving operational efficiency.

### Teaching Assistant - R, Statistics

Minneapolis, USA

Carlson School of Management, University of Minnesota

Aug 2023 – May 2024

- Facilitating hands-on usage of R, grading assignments, and communicating with the students.

## Skills

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**Programming Languages:** Python, Java, SQL, R, MATLAB, HTML, PLC

**Data Analysis:** ChatGPT, Excel, Pandas, Numpy, Scikit-Learn, Matplotlib

**Software and Tools:** AutoCAD, Git, Linux, Microsoft Office, Jupyter Notebook, APIs

**Soft Skills:** Communication, Analytical, Problem Solving, Agile, English (Fluent)

## Academic Project Experience

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### 3D Semantic Reconstruction: PyTorch, Scikit-Learn, UNet, COLMAP, kNN Classifier

Fall 2023

- Performed a 2D segmentation with an IoU of 0.75 and generated 3D points from images.
- Constructed a pipeline from 2D semantic labels and 3D reconstructed points to estimate 3D semantic reconstruction.

### Off-road Driveable Area Segmentation: PyTorch, CycleGAN

Fall'23

- Trained the CycleGAN model on the "Yamaha-CMU-Off-Road" dataset to extend it.
- Employed the model for generating ground truth labels with a Dice score of 0.56.

### Box Sorting using Baxter Robot: Robot Operating System (ROS), OpenCV, Computer Vision

Spring'23

- Programmed Baxter robot to pick up color-coded boxes and place them on corresponding stacks.

### Web App for FUnIE-GAN, Deep-SESR, SVAM: Python, flask, HTML, and CSS

Summer'23

- Created flask-based API and deployed the web app on the IRV lab server.

### Generalizability of FLAN-T5 Model Using Composite Task Prompting: Hugging Face, LLM, NLP

Spring'23

- Designed compositions of tasks to experiment with the generalizability of the model.
- Recognized the limitations caused by the prompt structure and several keywords.